When Should a Rule Learn? Predicting the Rule, LLM, and Trained-Model Composition for Classification Streams

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Abstract

A production classification site can be served by a hand-written rule, a large language model (LLM) call, or a task-trained model. Common practice treats the rule-to-LLM-to-ML progression as a fixed ladder whose destination is the trained model. We ask a different question. Given a stream's measurable characteristics, which of these tiers should serve it, and when? We score four tiers (keyword rule, zero-shot LLM, few-shot LLM, and a task-trained-model training curve) on a common test set across 19 public text datasets, with preliminary image and audio. Two early-observable properties, label cardinality and rule accuracy over chance, predict which tier wins: high-cardinality streams favor the task-trained model, while binary and low-cardinality streams favor the LLM. We operationalize this as Adaptive Graduated Autonomy (AGA), which selects the terminal tier per stream under a multi-objective policy (modality feasibility and a latency budget as hard constraints, accuracy within a significance band, operating cost minimized) and graduates between tiers using a statistical non-inferiority gate. The study measures the accuracy, cost, and latency axes; modality feasibility enters as a hard filter rather than a measured scalar. AGA matches the fixed-ladder baseline's accuracy while using about 44% fewer labeled outcomes, and it keeps the LLM as the terminal tier on roughly half of the streams. We report two negative results. A learned cross-site predictor of the best tier does not generalize at this number of datasets, so per-site measurement is required. And a vision-LLM-on-spectrogram proxy fails on environmental audio. The contribution is the characterization and the cost-reducing adaptive algorithm, not the cascade or the gate, which are prior art.

1 Introduction

Most production software contains many small classification decisions: route this ticket, tag this message, flag this content, pick this branch. Each is a function from an input to one of a fixed set of labels. Teams build these three ways, with a hand-written rule, an LLM call, or a task-trained model. The choice is usually made once per site, by intuition, and rarely revisited; there is no shared, evidence-based basis for it. Two progressions are documented in practice. The first replaces brittle rules with learned models, a migration long noted as a source of technical debt (Sculley et al. 2015; Breck et al. 2017). The second prototypes with an LLM and distills to a cheaper model to cut inference cost (FrugalGPT; LLM-as-teacher distillation). Both implicitly treat a trained model as the destination, an assumption also built into autonomy frameworks whose ceiling is a trained model. We test that assumption with data rather than assert a ladder.

For a given classification stream, which tier should serve it, and when? We treat this as a measurement problem. We score all three tiers (rule, LLM, task-trained model) on the same test set, on a shared axis of labeled outcomes consumed, across 19 text datasets spanning sentiment, intent, moderation, emotion, topic, and spam, with preliminary image and audio. The answer is not "always the trained model." It depends on measurable properties of the stream, chiefly label cardinality and how far the day-zero rule sits above chance. For a large fraction of streams the LLM is the right terminal tier, not a way-station.

We turn this into an algorithm, Adaptive Graduated Autonomy (AGA), that chooses the terminal tier per stream under a multi-objective policy (modality feasibility and a latency budget as hard constraints, accuracy within a significance band, operating cost minimized). It graduates between tiers using a non-inferiority gate, moving to the cheaper tier as soon as that tier ties, not only when it wins. Against the fixed "always graduate to ML" baseline, AGA matches accuracy while using far fewer labeled outcomes.

We do not claim the mechanism as novel. LLM cost-cascades (FrugalGPT) and learning-to-defer already route between models, and paired-McNemar comparison of classifiers is standard. The contributions are: (1) a characterization of which dataset properties predict the cost-optimal tier; (2) an algorithm that exploits it to cut labeled-data cost at equal accuracy; and (3) evidence on where a learned cross-site policy does and does not work, including two negative results.

1.1 Contributions

1. An aligned multi-tier benchmark. Rule, zero-shot LLM, few-shot LLM, and a task-trained-model curve, all scored on one common test set per dataset, on a shared labeled-outcomes axis, across 19 text datasets and preliminary image and audio.
2. A predictive characterization. Label cardinality (Spearman $\rho \approx -0.7$ against outcomes-to-graduate) and rule-over-chance predict which tier wins and how deep ML must train.
3. AGA, which chooses the terminal tier and graduates under a non-inferiority gate, matching fixed-ladder accuracy at about 44% fewer labeled outcomes.
4. A lifecycle-level safety bound. A cumulative Type-I bound over the whole graduation chain, not just one transition, with the rule as a persistent accuracy floor.
5. Two negative results. A learned cross-site tier-predictor does not generalize at 21 datasets, and spectrogram-vision fails on audio. We also give a tunable non-inferiority gate for cost-aware graduation.

2 Setup

Tiers. RULE, then the LLM (a pretrained model used zero- and few-shot, consuming no task labels), then a task-trained model (a classifier fit to the site's labeled outcomes, for example a logistic-regression head or a fine-tuned transformer). What separates the last two is training provenance, a borrowed general model versus a model fit to this task, not architecture. For brevity the tables abbreviate the task-trained tier as ML and the two LLM settings as LLM-zs (zero-shot) and LLM-fs (few-shot).

Datasets. 19 text datasets, plus CIFAR-10 (image) and ESC-50 (audio).

Aligned protocol. A common test subset per dataset (text n=400). The rule is built from a 100-example seed. ML is evaluated as a budget curve. The LLM (Claude Haiku) is run zero- and few-shot, with at most 40 exemplars. Each tier is placed at its true labeled-outcome cost: rule about 100, zero-shot 0, few-shot k, ML the training budget.

Two costs. We separate the one-time labeled-outcome cost (to train ML or seed the rule) from the perpetual per-call inference cost (the LLM bills every call; rule and trained ML are roughly free per call). AGA's savings are reported on the labeled-outcome axis. The steady-state inference cost is what makes graduating off the LLM valuable.

3 The composition varies by stream

Sorted by label cardinality, a gradient appears: high cardinality favors trained ML, while binary and low-cardinality favor the LLM. Two rows are artifacts addressed in Section 7. ESC-50 ML "wins" only because the spectrogram-vision proxy fails, and CIFAR-10's LLM "win" reflects the pixel-logreg baseline; a frozen-ViT baseline reaches 0.95 and beats the LLM's 0.78, so image in fact favors ML.

dataset	labels	rule	ML	LLM-zs	LLM-fs	best
clinc150	151	0.172	0.838	0.748	0.725	ML
banking77	77	0.225	0.870	0.777	0.807	ML

dataset	labels	rule	ML	LLM-zs	LLM-fs	best
hwu64	64	0.247	0.805	0.850	0.865	LLM-fs
esc50	50	0.058	0.283	0.017	0.050	ML
atis	26	0.733	0.848	0.598	0.815	ML
twenty_newsgroups	20	0.133	0.660	0.685	0.680	LLM-zs
dbpedia14	14	0.632	0.968	0.968	0.978	LLM-fs
codelangs	12	0.892	0.986	1.000	1.000	LLM-zs
cifar10	10	0.173	0.327	0.780	0.340	LLM-zs
yahoo_answers	10	0.240	0.670	0.740	0.720	LLM-zs
snips	7	0.755	0.983	0.970	0.983	ML
emotion	6	0.323	0.818	0.632	0.608	ML
trec6	6	0.420	0.855	0.795	0.887	LLM-fs
ag_news	4	0.375	0.895	0.853	0.892	ML
tweet_emotion	4	0.403	0.613	0.828	0.810	LLM-zs
imdb	2	0.532	0.853	0.938	0.958	LLM-fs
rotten_tomatoes	2	0.487	0.723	0.930	0.940	LLM-fs
sms_spam	2	0.920	0.945	0.887	0.988	LLM-fs
sst2	2	0.512	0.770	0.968	0.978	LLM-fs
tweet_hate	2	0.497	0.560	0.770	0.640	LLM-zs
tweet_offensive	2	0.660	0.775	0.752	0.730	ML

4 Adaptive Graduated Autonomy

The terminal-tier selection is multi-objective, applied in priority order. First, modality feasibility, a hard filter (a spectrogram image cannot convey words, for instance). Second, the latency budget, a hard constraint; a tier that exceeds the real-time budget is removed. Third, accuracy, where tiers within a Wilson-CI significance band of the best feasible tier are retained. Fourth, operating cost, minimized within that band, counting perpetual inference cost and labeled cost. The LLM is chosen only when it is significantly more accurate and feasible under the latency and modality constraints; otherwise a rule or trained-ML tie graduates off the LLM. Cost is the axis this study measures. Latency and modality are first-class in the policy and default to unconstrained.

ML graduates once it ties the LLM under the non-inferiority gate, not once it beats it. The margin ϵ is operator-tunable. The router is permanent: it holds the rule floor, watches for drift, and re-escalates to the LLM when the data calls for it. What shrinks is the dependence on the LLM, not the router.

4.1 A lifecycle-level safety guarantee

The per-transition gate gives the standard guarantee that advancing to a worse-than-current tier has probability at most , the McNemar Type-I error. That part is not novel; it is the test's definition applied to a promotion decision. The lifecycle-level statement is what separates AGA from a single gated comparison.

Proposition (cumulative safety). Consider a graduation trajectory of at most m gated transitions, each evaluated by a paired test at level with non-inferiority margin ϵ. With probability at least $1 - $ m, every advance is to a tier non-inferior (within ϵ) to the current one. Consequently the deployed accuracy stays within ϵ of the best previously certified tier, and because the rule is a persistent fallback, operating accuracy is bounded below by (rule accuracy $- \epsilon$) throughout, for any m.

Proof sketch. Each gated advance to an inferior tier is a Type-I event with probability at most . A union bound over at most m transitions bounds the probability of any such event by m. Absent any Type-I event, each advance is within-ϵ non-inferior, so by induction the deployed accuracy stays within ϵ of the best certified tier, and the rule floor lower-bounds it.

Remarks. The union bound is loose, and independence across transitions is an idealization, since the tests reuse the accumulating stream. If many transitions are expected, should be spent with a correction such as /m. The guarantee is on certified accuracy under the test's assumptions, not a distribution-free promise.

What it adds over a single McNemar test is a composable safety statement for a multi-tier lifecycle, which is what a practitioner needs to let the system graduate unattended.

4.2 Measured latency and cost

Per-call latency (p50): rule 0.003 ms, task-trained model 0.1 ms, MiniLM-embedding ML 6 to 23 ms, LLM 530 to 570 ms. The rule is roughly 200,000 times faster than the LLM, and task-trained model about 5,000 times. LLM per-call cost is $0.0001 to $0.0006 and scales with the length of the label prompt; rule and ML are near zero. Under a real-time budget (for example 50 ms) the LLM is excluded by construction, so latency forces graduation to local tiers. See latency_profile.json.

4.3 AGA versus the fixed ladder

AGA matches accuracy at much lower labeled-outcome cost. In the raw output below, model_zeroshot and model_fewshot denote the LLM tier (zero- and few-shot) and ml denotes the task-trained tier.

dataset	oracle	AGA(LODO)	AGA_acc	fixed_acc	AGA_cost	fix_cost
ag_news	ml	model_zeroshot	0.853	0.895	0	5000
atis	ml	ml	0.848	0.848	1000	1000
banking77	ml	model_fewshot	0.807	0.870	40	5000
cifar10	model_zeroshot	ml	0.327	0.327	1000	1000
clinc150	ml	ml	0.838	0.838	5000	5000
codelangs	ml	ml	0.986	0.986	500	500
dbpedia14	ml	ml	0.968	0.968	10000	10000
emotion	ml	model_zeroshot	0.632	0.818	0	5000
esc50	ml	model_fewshot	0.050	0.283	8	100
hwu64	model_fewshot	ml	0.805	0.805	5000	5000
imdb	model_fewshot	model_fewshot	0.958	0.853	40	5000
rotten_tomatoes	model_fewshot	model_zeroshot	0.930	0.723	0	5000
sms_spam	model_fewshot	ml	0.945	0.945	2000	2000
snips	ml	ml	0.983	0.983	500	500
sst2	model_fewshot	model_zeroshot	0.968	0.770	0	5000
trec6	ml	model_zeroshot	0.795	0.855	0	5000
tweet_emotion	model_zeroshot	ml	0.613	0.613	2000	2000
tweet_hate	model_zeroshot	model_fewshot	0.640	0.560	40	250
tweet_offensive	ml	model_fewshot	0.730	0.775	40	2000
twenty_newsgroups	ml	ml	0.660	0.660	10000	10000
yahoo_answers	model_zeroshot	ml	0.670	0.670	10000	10000

```
LODO oracle-match: 7/21 = 0.33
Mean accuracy: AGA 0.762 vs fixed-ladder (ML_PRIMARY) 0.764
Mean labeled outcomes: AGA 2246 vs fixed-ladder 4017
AGA keeps LLM as terminal (skips ML training): 10/21
AGA picks rule/ML over a not-better LLM (avoids perpetual cost): 6/21
```

4.4 Negative result: cross-site prediction does not yet generalize

A gradient-boosted predictor of the best tier, trained on the early-observable characteristics, scores below the majority baseline under leave-one-dataset-out at 21 datasets. Its apparent skill also moves with test-set noise, swinging from 0.62 to 0.41 between n=200 and n=400 before significance-aware labeling. We conclude that per-site measured routing is what works today, and that a learned cross-site predictor needs a much larger collection of streams. The practical reading is to measure rather than guess.

5 Preliminary results on image and audio

The gate is modality-agnostic, since it consumes (decision, outcome) pairs; only the LLM input path is modality-specific. The rule and ML tiers transfer to CIFAR-10 and ESC-50. The LLM tier is preliminary. Claude vision on CIFAR images reaches 0.78 zero-shot, though the pixel-logreg ML used in the aligned table is a weak baseline (see Section 7). A spectrogram-to-vision proxy for ESC-50 fails, scoring 0.017, below chance, so a native-audio model is needed. We treat these as preliminary evidence that the lifecycle generalizes across modalities, not as headline results.

6 Practical implications

For a team choosing how to serve a classification stream, the results suggest four guidelines.

1. The trained model is not the default destination. On many streams, especially binary and low-cardinality ones, a pretrained LLM is the most accurate tier and is the right place to stop; do not assume the goal is to graduate to a task-trained model.
2. Check two cheap signals first. Label cardinality and how far a 100-example rule sits above chance predict which tier will win, before any training. High cardinality with a near-chance rule points to the task-trained tier; a binary task with a weak rule points to the LLM.
3. Graduate on a tie, not a win. The rule and a trained model are far cheaper to operate than an LLM (microseconds to milliseconds versus roughly half a second per call here), so moving to the cheaper tier as soon as it is statistically non-inferior captures most of the cost and latency benefit at no accuracy loss.
4. Measure per site rather than predict across sites. A learned predictor of the best tier from stream characteristics did not generalize at 21 datasets, so the reliable approach is to instrument each site and let the gate decide, not to guess from a model of the choice.

7 Limitations

The task-trained tier defaults to the cheap day-N head (TF-IDF, pixels, or MFCC features with logistic regression). To check whether the “LLM wins” results are an artifact of a weak baseline, we ran two stronger baselines. Frozen sentence-transformer embeddings (all-MiniLM-L6-v2) with logistic regression still lose to the LLM on short-text sentiment and moderation (sst2 0.80 vs 0.98; rotten_tomatoes 0.74 vs 0.94; tweet_hate 0.53 vs 0.77). A fine-tuned DistilBERT (4k examples, 3 epochs) narrows the gap but does not close it (sst2 0.89 vs 0.98; imdb 0.87 vs 0.96, about 9 points). On high-cardinality intent, a tuned TF-IDF and logistic regression remained the strongest classical baseline (banking77 0.87, above fine-tuned DistilBERT’s 0.80 at this budget), consistent with ML winning there. The central finding therefore holds across baseline strength. Two caveats apply: the fine-tune was light and untuned, so heavier tuning could narrow but is unlikely to erase the sentiment gap, and frozen embeddings are not uniformly stronger (imdb MiniLM 0.78 is below TF-IDF’s 0.85 because of truncation). For image, a real baseline resolves the ambiguity: frozen ViT embeddings with logistic regression reach 0.95 on CIFAR-10, above the zero-shot vision LLM’s 0.78, so CIFAR’s apparent “LLM win” was the pixel-logreg strawman and image favors ML, as high-cardinality text does. The aligned table still lists the pixel-logreg tier; the ViT number is the robustness check.

Other limitations. Latency and cost are measured on a sample (latency_profile.json), and LLM latency is network-bound and will vary. We use a single LLM (Claude Haiku) and a single prompt, so accuracies are sensitive to prompt and model. The multimodal LLM tier is a proxy (spectrogram or sampled frames); native audio and video models are future work, and spectrogram-vision is shown to fail on environmental audio. The cross-site predictor does not generalize at this number of datasets (Section 4.4). With n=400 test subsets the accuracy confidence intervals are about ś0.04, and we use significance-aware tie bands accordingly.

8 Related work

FrugalGPT and related LLM cascades route for cost. Learning-to-defer routes per example. Champion-challenger and shadow deployment are standard MLOps. The paired-McNemar test follows Dietterich. AGA differs by choosing the terminal tier per stream from measurable characteristics and graduating off the LLM under a non-inferiority gate, rather than routing per example or hand-tuning a cascade.

9 Reproducibility

A single command, `scripts/reproduce_dmlr.sh`, regenerates every number. The MODEL stages skip automatically without an API key. See REPRODUCE.md.